

You are in an elevator with \$2.

Every minute a chicken appears

Every minute in the elevator your money doubles.

If a chicken touches you, you lose all the money.

You can leave whenever.

CS 340

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Q1

~Code~
340



How long will you try for?

Building Blocks 0b10
(Selection and Information Storage)



Updates

1. MP 1.1, MP 1.2, MP 1.3

1. Environment check-off
a. Check your PL grade

1. HW 3

1. Regrades - report an error on PL

→ Sept 15
→ ~~16~~

wed

HW1 HW2
graded

The screenshot shows a question interface for "Question HW2.8". The total points are 25. Below the question title, there is a section for "Manually-graded question" which includes a button to "Report an error in this question". A red circle highlights this section. Inside the circle, there is a text area to "Describe the error in this question" and a yellow "Report error" button at the bottom right. Red arrows point to the text area and the button. The background of the page shows parts of other questions, such as "malloc with a", "happens, you can", "contents. In C++ a", "to be compiled with", "capacity before", "failure.", and "ccess, 0 on failure."

Building Blocks Ob10

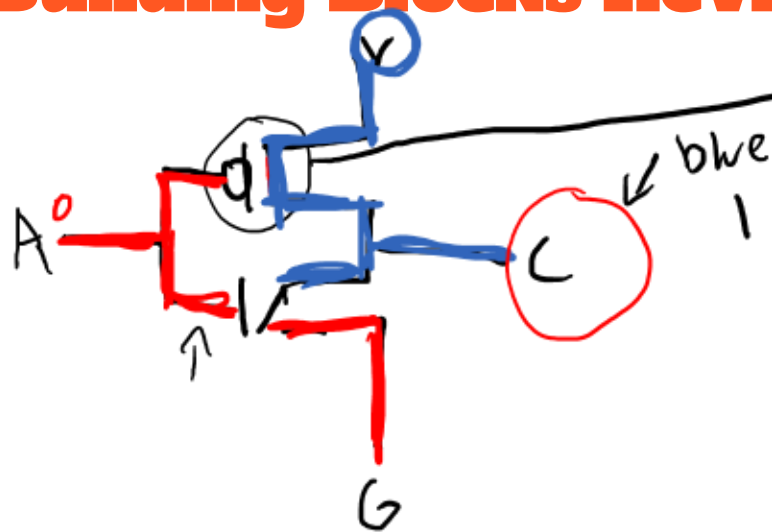
Today's LGs:

- See that you can use gates to build math and selection.
- Have a brief understanding of why we use base-2 for storage
- Be able to articulate that
 - Computers have different types of storage hardware
 - We utilize the different types of storage by using caching
 - Caching algorithms rely on spatial and temporal locality
- Be able to identify if code is cache friendly or not

Agenda

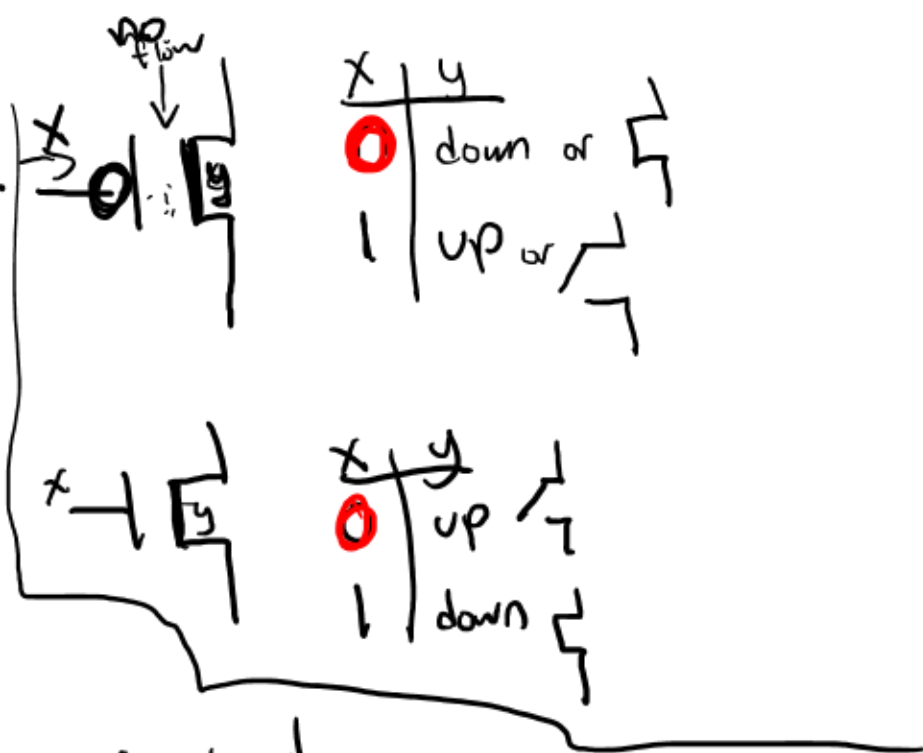
1. Arithmetic Calculations
2. Selection
3. Storage

Building Blocks Review



V - blue water (1)

G - red water (0)



purple is bad

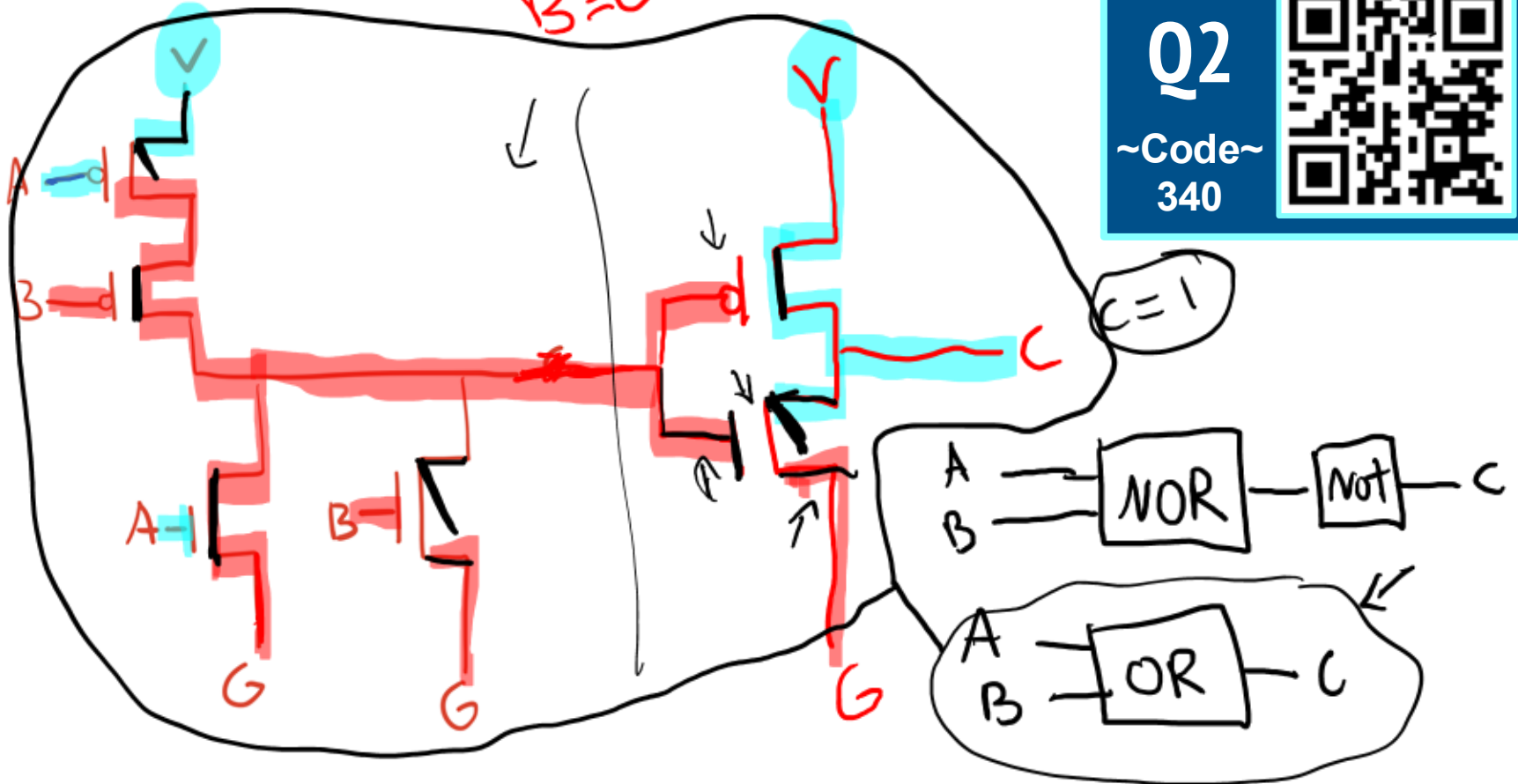
Example Gate

A=1 blue
B=0 red

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Q2

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Building Blocks Review

↑ ↓ voltage is used with transistors to implement logic

↑ ↑
1 0

Binary is 1, 0 so it works with ↑ ↓ voltage

Build on this...

Arithmetic Calculations

What is $0b011 + 0b001$ in decimal?

$$3 + 1 = 4$$

$$0b100 = 4$$

- A) 100
- B) 3
- ☒ C) 4
- D) 8

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Q3

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X + Y = Z in Logic

AND = $\&$ XOR = $\wedge$ or = $|$ not = $\neg$

$$3 + 1 = 4$$

$$x = 3 = 0b011$$

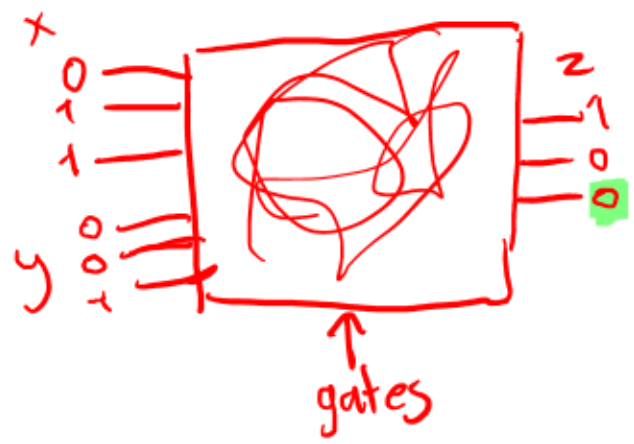
$$y = 1 = 0b001$$

$$x = x_2 x_1 x_0 = 0 \overset{1}{\downarrow} 1 \overset{1}{\downarrow}$$

$$y = y_2 y_1 y_0 = 0 0 \overset{1}{\downarrow}$$

$$z = z_2 z_1 z_0$$

$$\begin{array}{r} 011 \\ + 001 \\ \hline 100 \end{array}$$



$$z_0 = x_0 \wedge y_0 = 1 \wedge 1 = 0$$

$$z_1 = c_1 \wedge x_1 \wedge y_1 = (1 \wedge 1) \wedge 0 = 0$$

$$c_1 = x_0 \& y_0 = 1 \& 1 = 1$$

$$c_2 = \dots = 1$$

$$z_2 = c_2 \wedge x_2 \wedge y_2 = (1 \wedge 0) \wedge 0 = 1$$

Addition Formulas

$$Z_0 = x_0 \oplus y_0$$

$$C_1 = x_0 \& y_0$$

$$Z_i = C_i \oplus x_i \oplus y_i$$

$$C_{i+1} = (x_i \& y_i) \vee (C_i \& (x_i \oplus y_i))$$

XOR, AND, OR

↓
make them
with
gates

Which diagram would build the logic in the bold box?

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$$Z_1 = x_1 \oplus y_1$$
$$C_1 = x_1 \& y_1$$

B



a	b	c
1	0	0
0	1	0
0	0	0
1	1	1

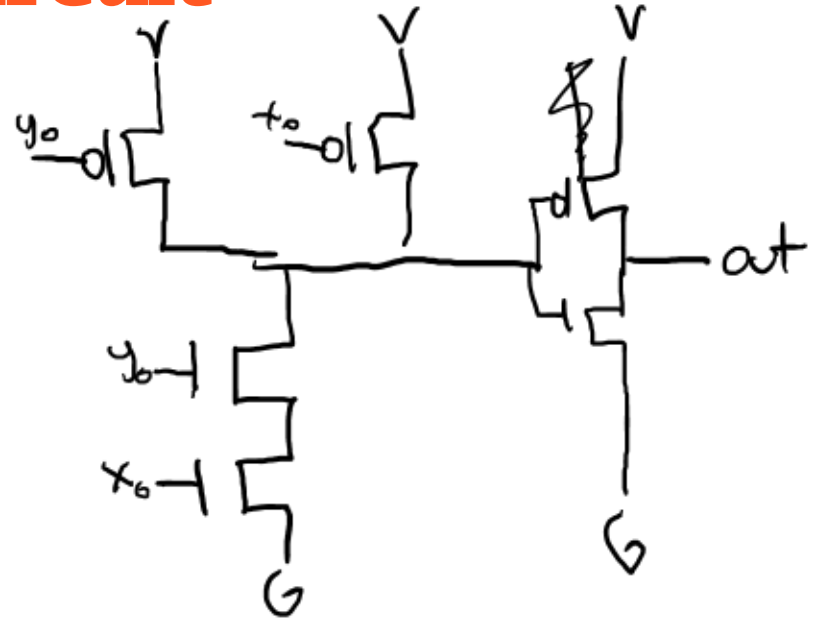
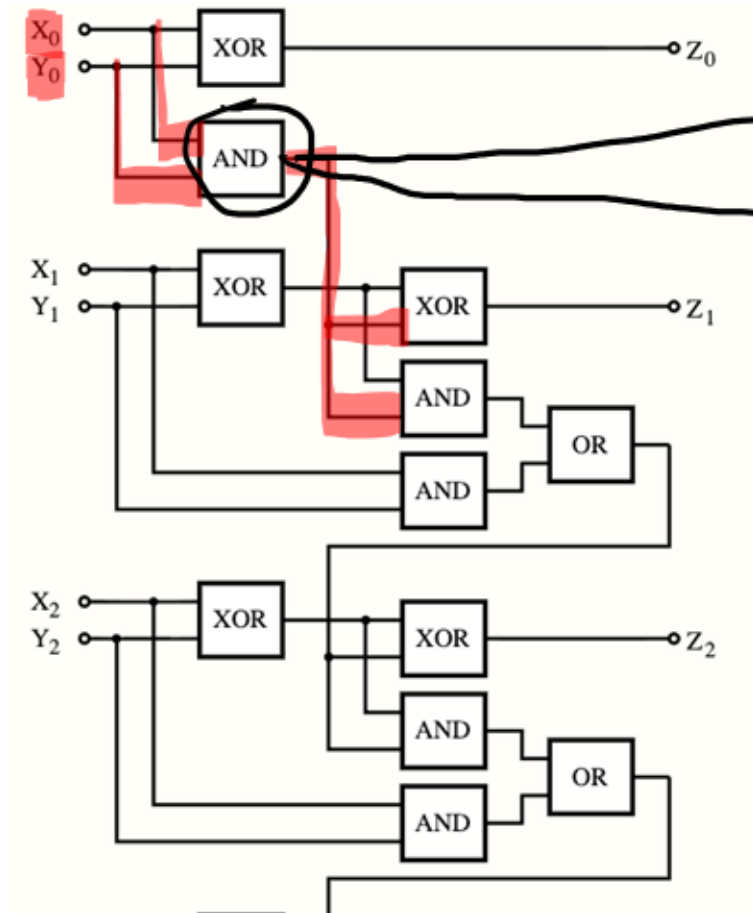
A



C



Ripple-carry adder circuit

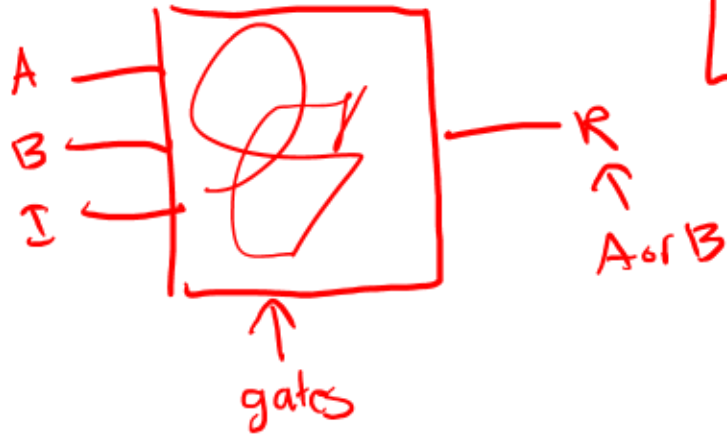


Selection

MUX (sounds like ducks with an m)

input

A, B, I [A, B]



A	B	I	MUX
0	0	0	A
0	0	1	B

2-MUX

$$R = (\neg I \& A) \vee (I \& B)$$

$$\begin{matrix} [1, 0] \\ A \quad B \end{matrix} \quad I=0$$

$$(1 \& 1) \vee (0 \& 0) \\ 1 \vee 0 = 1$$

What is the output when I is 1?

$[A, B]$
I = 1

$$R = (\neg I \& A) \vee (I \& B)$$
$$(0 \& A) \vee (1 \& B)$$

$$0 \vee (B)$$

B

- A) A
- ☒ B) B
- C) I

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A MUX can scale!

$$\left[\underset{A}{2}, \underset{B}{1} \right] \quad I = 0$$

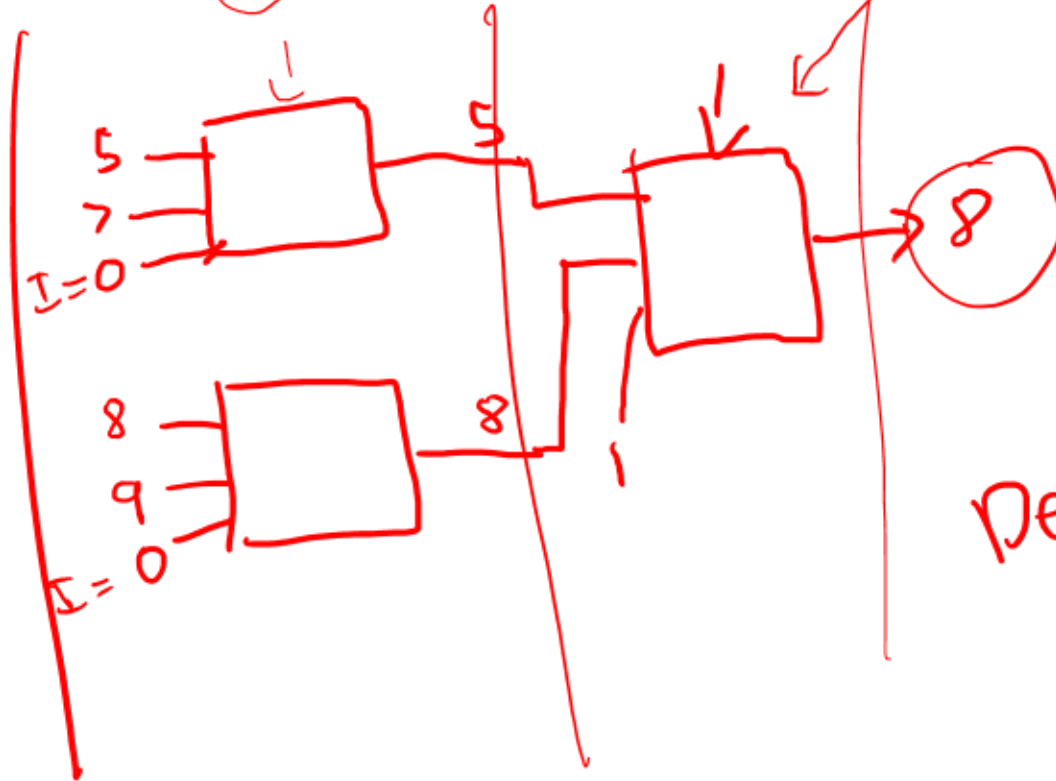
$$[0b10, 0b01] \quad I = 0b00$$

$$\begin{array}{r} \boxed{(\neg I \& A)} \mid (I \& B) \\ \begin{array}{r} 0b11 \\ \& 0b10 \\ \hline 0b10 \end{array} \quad \begin{array}{r} 0b00 \\ \& 0b01 \\ \hline 0b00 \end{array} \end{array}$$

$$\begin{array}{l} \rightarrow 0b10 \\ \text{or } 0b00 \\ \hline 0b10 = 2 \\ R = \end{array}$$

A MUX can scale in other ways too!

$[5, 7, 8, 9]$ $I = 2 = 0010$
0 1 2 3



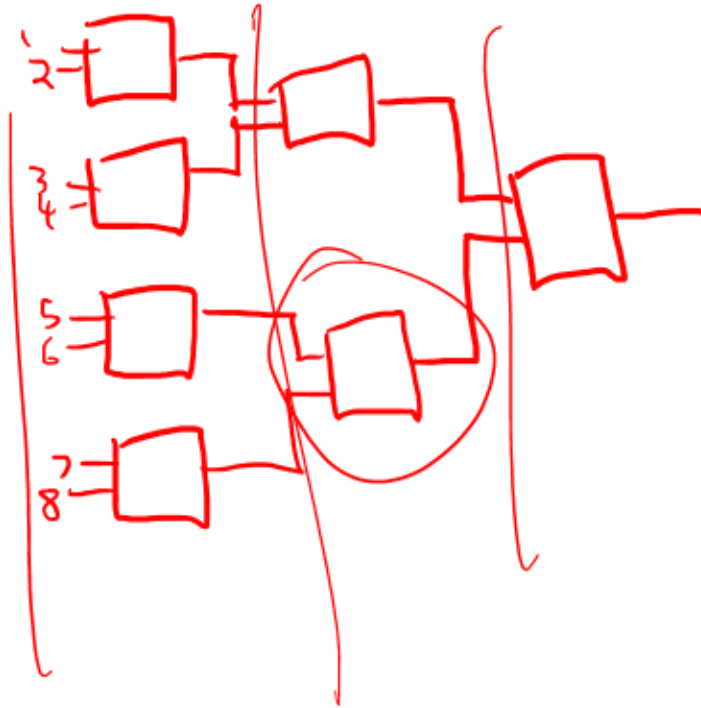
$\log_2(4)$

Depth 2

What depth of 2-MUX for a 8 selection?

$[1, 2, 3, 4, 5, 6, 7, 8]$ $I = 0-7$

- A) 1
- B) 2
- C) 3
- D) 4
- E) 5



$= 3$

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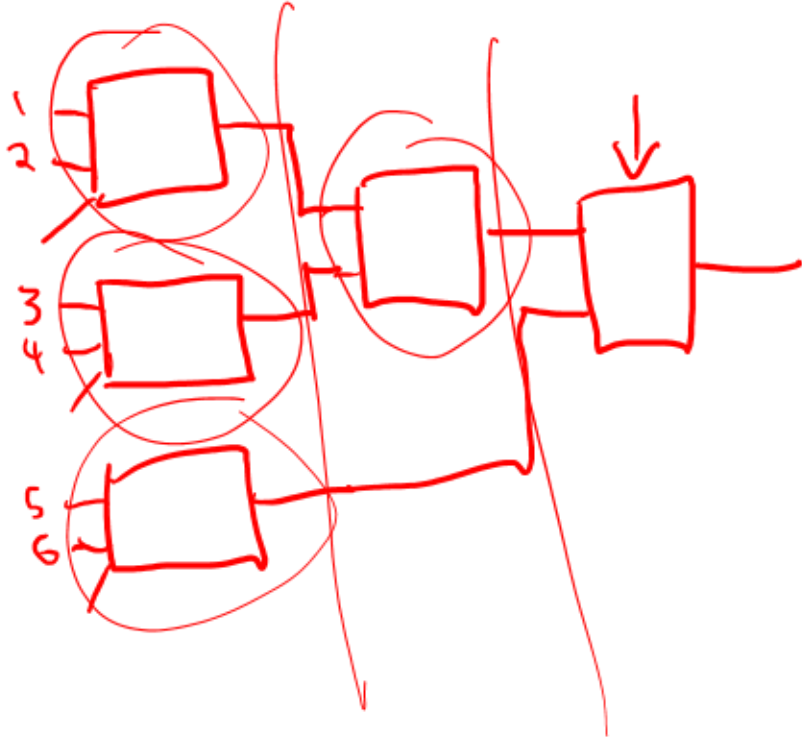
Q6

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What depth of 2-MUX's for a 6 input selection?

[1, 2, 3, 4, 5, 6]



- A) 1
- B) 2
- C) 3
- D) 4
- E) 5

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Q7

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3

MUX Takeaways

2-Mux is hardware that selects one of two

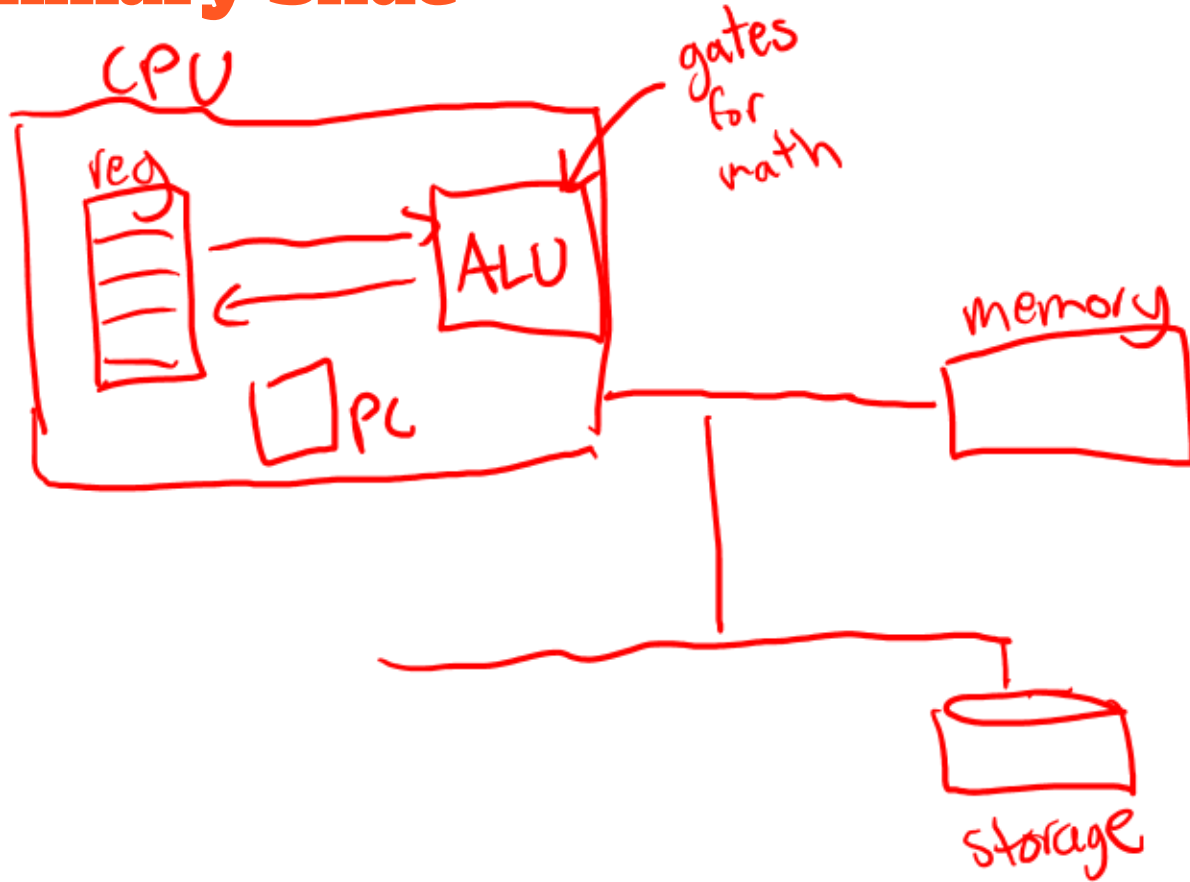
more depth == slower

rounded
up $(\log_2 \text{Size})$

selecting from 6 is same speed as selecting
from 8

$\Rightarrow$ why data is stored in powers of 2
~~128~~ 1286B

Summary Slide



Storage

Types of Storage

We store $\uparrow \downarrow$ voltages

gates, capacitors,

fast, slow

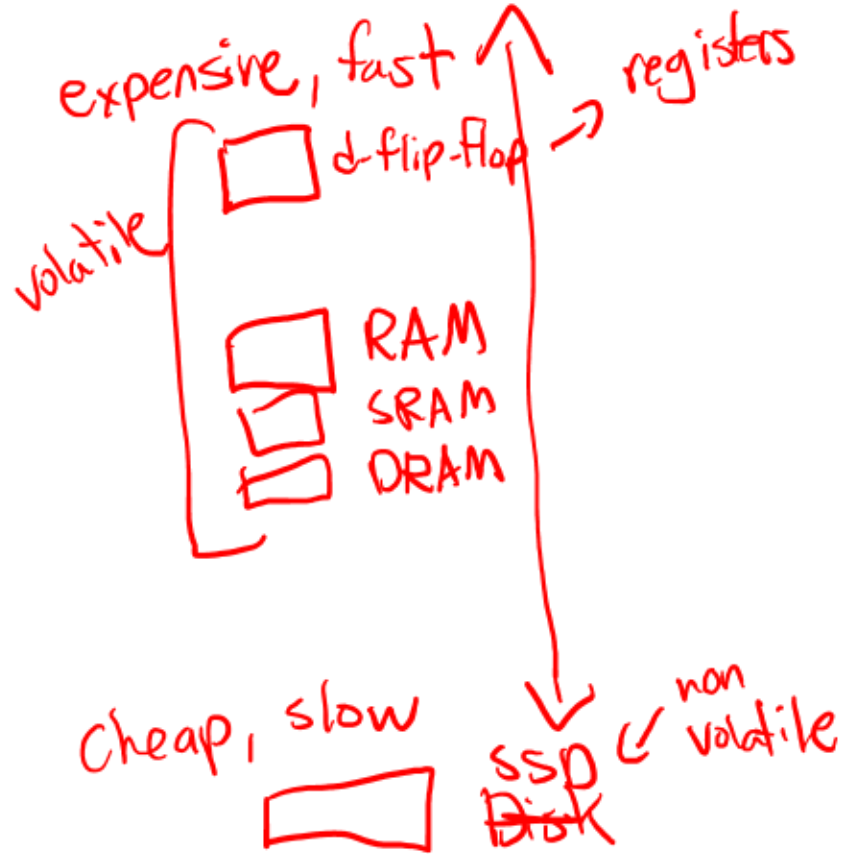
expensive, cheap $\leftarrow$ space

Volatile, non-volatile

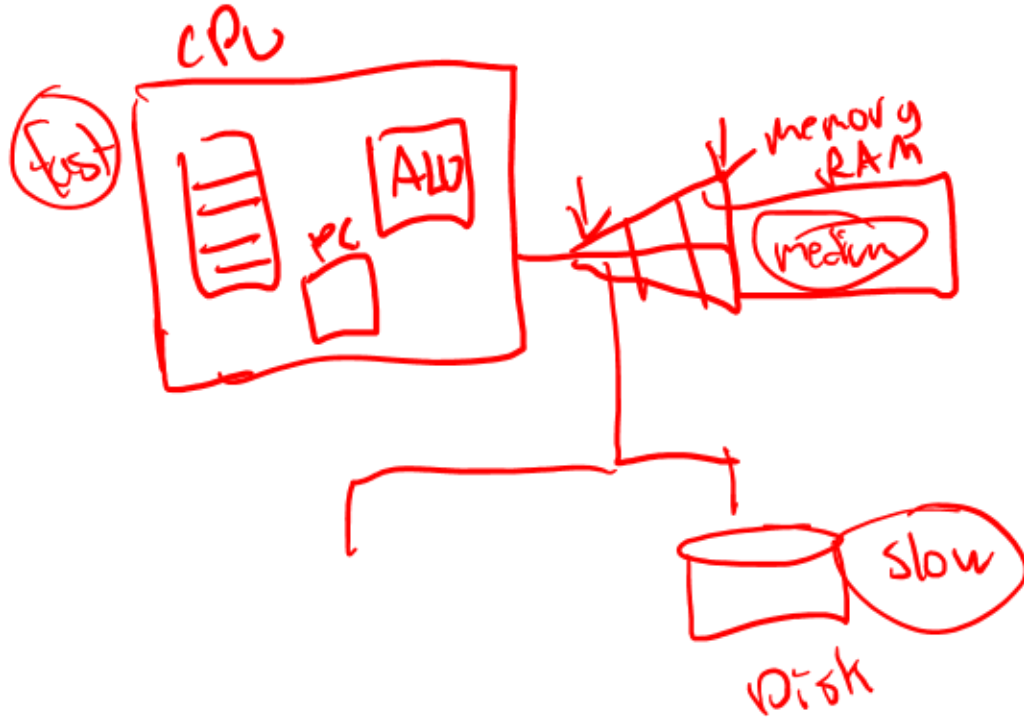
$\uparrow$
persist

only fast $\rightarrow$ too expensive

only cheap $\rightarrow$ too slow



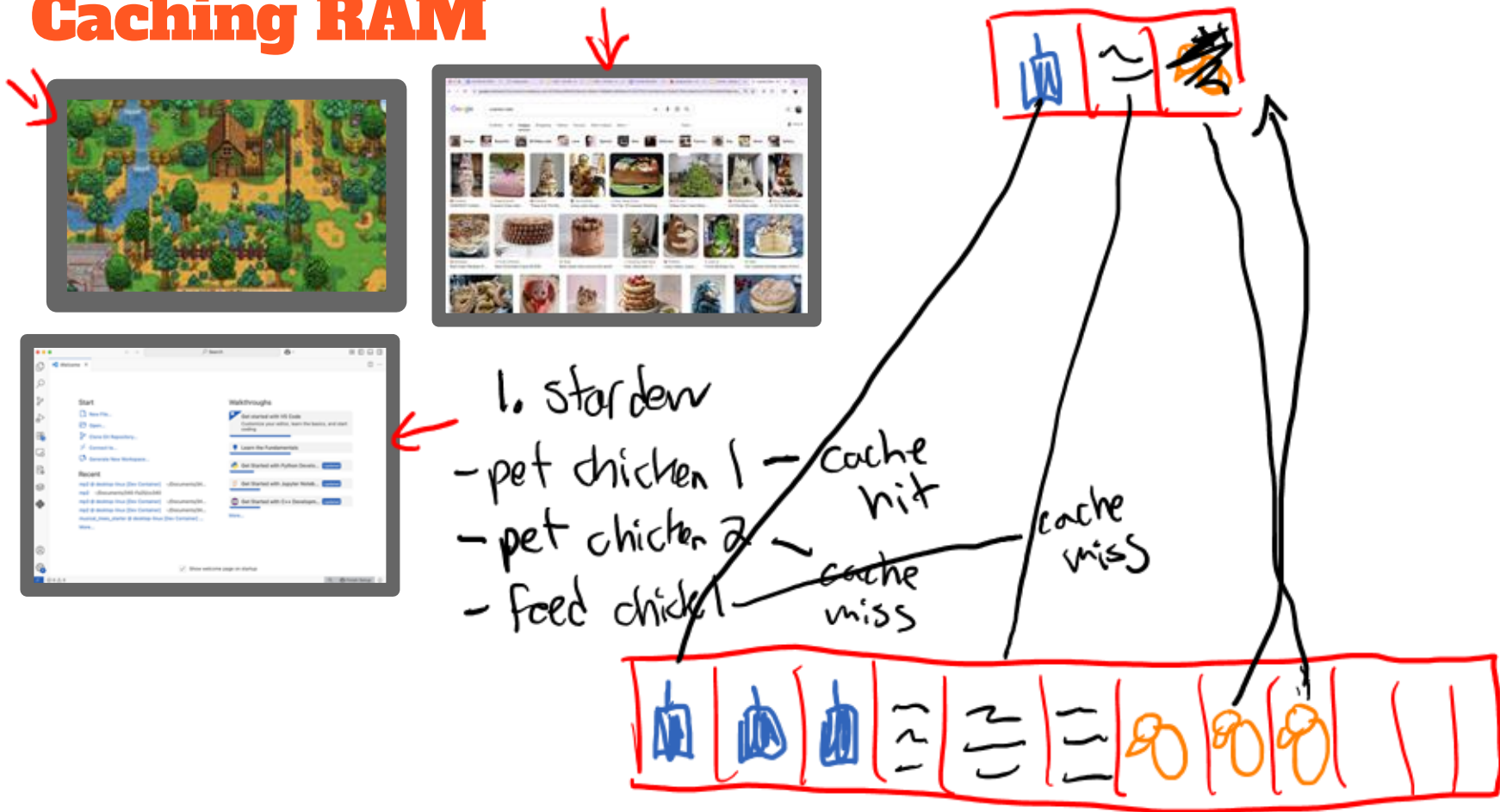
Hardware for Storing Information



how do we
use all these
types together

caching

Caching RAM



Caching - an algorithm for utilizing fast small memory and slow big memory.

Locality - the idea that computers often use nearby and similar information sequentially.

Spatial locality - things nearby are more likely to be used

Temporal locality - things used recently are more likely to be used again

Library Example

10 books



100 books



1,000 books



100,000 books



I want a book

- table - cache miss
- cart - cache miss
- shelf - cache miss
- archive - hit

spatial

- grab all 8 books in the series

- leave the book on the table

temporal

Computer Information Storage

How can you change this code for better locality?

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Q8

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```
6   int arr1[500];
7   int arr2[500];
8   //add stuff to arrays
9   int count = 0;
10  → for(int i = 0 ; i < 500; i++){
11      → if(arr1[i]%2 == 0) count++;
12      → if(arr2[i]%2 == 0) count++;
13  }
```

for (...) {

if (arr2[i]%2 == 0) count++;
slow
}



How can you change this code for better locality?

```
8   int doub[500][450];
9   //add stuff to doub
10  for(int col = 0; col < 450; col++){
11      for(int row = 0; row < 500; row++){
12          doub[row][col]++;
13      }
14  }
```

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Q9

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